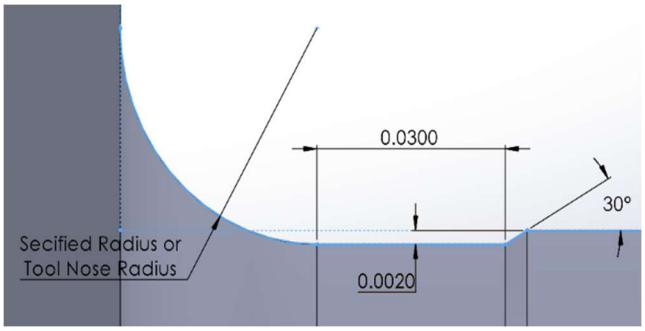
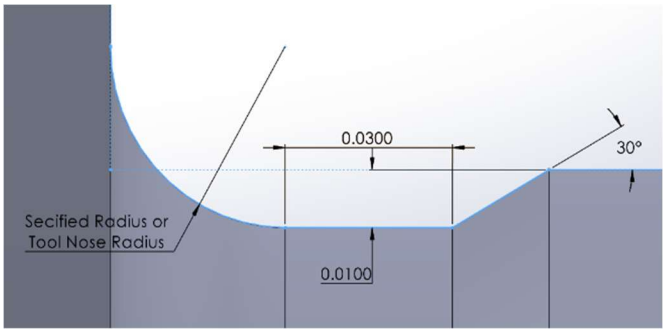
Standard Modeling Procedures

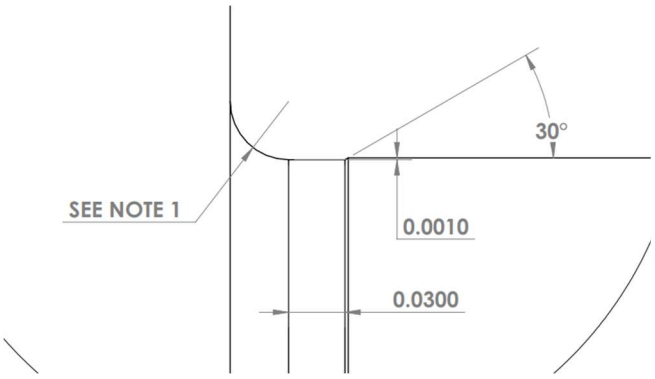
1. When starting a new model, create shaft **sketch** on Solidwork's **"Front"** plane. (Solidwork's front plane corresponds with Mastercam's Top plane.)
2. Make **centerline** of sketch and **one end** of shaft **coincident** to **origin**.
3. **Threads**
   a. Threads should be modeled to the mean of the thread major diameter unless specified on print.
   b. Chamfers on front of threads should be made to thread depth as a minimum.
   c. Chamfers on the back of threads should be 38° to the undercut diameter when allowed, for cutting with finish tool. (There are occasions when thread undercuts must be cut with groove tool.)
   d. **Thread Reliefs:**
      i. If a thread relief is not specified, the standard practice is to make the diameter of the relief .015" total under the spline minor, and make the width at least 2.5x the TPI.
4. **Undercuts (Disappearing)**
   a. Undercuts should be modeled as follows unless otherwise specified on print…



Standard Unspecified



Siemens Unspecified



ABB/Powertech Unspecified

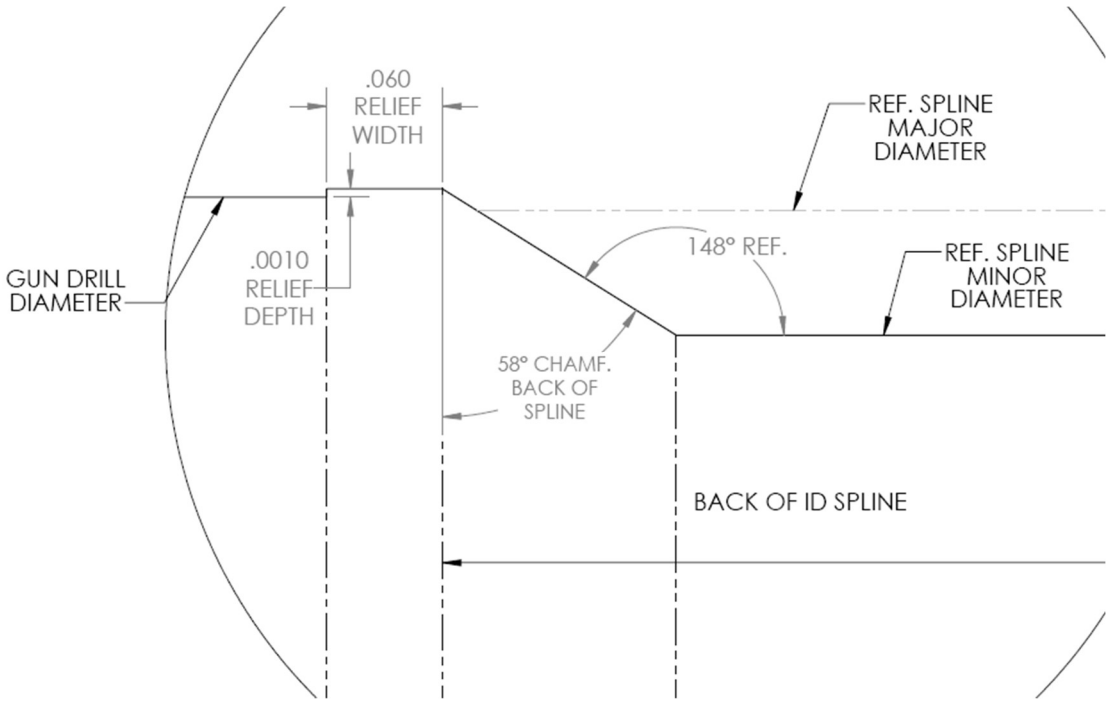
5. **Edge Breaks** should be modeled as follows…
   a. Break all edges before adding grind stock.
   b. Edge breaks should be reasonable depending on size of shaft, but unless otherwise specified be .010"-.030" on shoulders, .005" on grooves.
   c. Make sure not to break edges where "sharp" edges are called out.
   d. Break edges on front of threads x 45 deg., and break relief of threads x 38 deg. angle to allow cutting with finish tool. (Unless thread reliefs are specified.)
   e. Occasionally, some edges may need to be broken after grind stock is added. This may happen when there is no shoulder present and only a radius. These will need to be made using the chamfer feature.
6. **End work**

a. If shaft is being ran of face driven lathe, end work does not need to be modeled. The best practice would be to create them anyways in case we run them on a different machine later.
b. End work can be modeled in original shaft or modeled in second sketch then revolved cut.
c. Standard end work features can also be dragged in from design library features when available.

7. **Grind Stock and Grind Diameters**
   a. **Grind diameters** should typically be modeled to high side of tolerance. (There will be occasions when this does not work, such as when modeling to the high would remove the undercut on some Siemens prints.)
   b. **Grind stock** should be added with **"Move Face"** tool, and faces should be offset by amount below.

MIN. LENGTH	MAX LENGTH	GRIND STOCK ø
0	15.0	0.01
15.0001	30.0	0.012
30.0001	60.0	0.015
60.0001	-	0.020

      i. Sometimes "move face" does not work and will remove edge chamfers/fillets. In these instances, grind stock can be drawn manually and revolved.
   c. Specific for **ABB splines**…
      i. **External Splines** should be treated as a normal grind diameter. Diameter should be modeled to the high of the spline major, and then offset by .010" using the move face tool.
      ii. **Internal splines** should be modeled to the mean of the spline minor diameter.
      iii. Anytime there is a spline, 2 **tooling diameters** need to be identified for chucking and rolling on. These diameters should get **.010"** stock per side instead of the standard .0075" to be rough ground before splining. (This was previously .015" per side but Mike changed the standard on 6/12/2025)
      iv. ID splines will need a relief as follows:



8. **Modeling Stock**
    a. Modeling stock is helpful if the starting stock will not be a perfect solid cylinder. (Gun drill, in process models, assemblies.)
    b. If a stock model is needed, it is best to do so in the current model on a different configuration called “Stock”.
9. **Configurations**
    a. Model configurations should be kept to a minimum to reduce the time it takes to save and regenerate a model.
    b. Configurations can be made for drawings that contain part number tables where only a few dimensions change. In these instances, the use of a configuration table is helpful.
    c. Configurations can also be made for a part that has multiple steps in the manufacturing process that require part to be turned to different dimensions.
10. **Parts With Mating Components and Stock**
    a. Parts with multiple prints/details that get assembled before turning final shaft should be made as independent models in the drawing folder and can be assembled in an assembly to be used as stock in Mastercam.
11. **ABB Specific Notes**
    a. Whenever shafts go to heat treat, any internal diameters with a tolerance of .002 or tighter will need .010” (.020” total) for cleanup after heat treat.
    b. Nuclear shafts and explosion proof surfaces are critical.
    c. Explosion proof surfaces where the finish requirement is not listed do not need to be ground. In the past, we have ground them but Mike decided (8/22/2023) that was not necessary, and if we run across any like that, they need to be updated to turn to size.
12. **Saving Models/Programs**
    a. Models should be saved in the following location: G:\Manufacturing\Programming\CAM Files\”Customer Name Folder”
    b. Folder should be named drawing number, and model should be named drawing number.
    c. When there is a new revision, old revision model should be placed in a folder with old revision number and new model should be placed in the drawing folder